Yoann PULL

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Current Position

- **PhD Candidate in Economics** *October 2024 – Present*
  University of Orléans, Laboratoire d'Économie d'Orléans Orléans, France
  - Thesis title: *Robust Econometric Methods for Credit Risk Model Validation, Monitoring, and Stress Testing*.
  - Supervisor: Professor Christophe Hurlin. Expected defence: late 2027.
- **Research Scientist** *September 2024 – Present*
  Square Management Paris, France
  - Industry partner for the CIFRE PhD programme.
  - Conduct research and consulting assignments for banking institutions on credit risk modelling, probability of default calibration, model validation, and stress testing.
  - Develop econometric, Bayesian, and sequential testing methods for financial risk management and credit portfolio monitoring.
  - Design and deliver training sessions for banking and insurance clients on the impact of climate risk on financial risks.

Research Interests

Financial econometrics, credit risk modelling and model validation, stress testing; Bayesian and sequential inference, time series and forecasting, climate risk.

Visiting Positions

- **Visiting Student Researcher** *September – December 2026*
  Department of Statistics, Stanford University Stanford, CA, United States
  - Host: Professor Aaditya Ramdas. Research on sequential and anytime-valid inference.

Education

- **PhD in Economics** *2024 – Present*
  Laboratoire d'Économie d'Orléans, University of Orléans Orléans, France
- **M.Sc. in Applied Mathematics and Statistics, Financial Engineering Track** *2022 – 2024*
  University of Bordeaux Bordeaux, France
  - Main fields: probability, statistics, applied mathematics, financial econometrics, quantitative finance, risk management, time series, machine learning, and credit risk.
- **B.Sc. in Pure Mathematics** *2019 – 2022*
  University of Bordeaux Bordeaux, France
  - Main fields: measure theory, probability, statistics, optimization, linear algebra, and abstract algebra.
- **First Year of B.Sc. in Mathematics** *2018 – 2019*
  University of French Polynesia Punaauia, Tahiti, French Polynesia

Funding

- **CIFRE Doctoral Fellowship** *2024 – 2028*
  ANRT, Square Management, University of Orléans Industry–academic doctoral funding
  - Doctoral funding awarded under the French CIFRE programme for an industry–academic research project on credit risk model validation, monitoring, and stress testing.

Publications

- **A Bayesian Approach to Probability of Default Model Calibration: Theoretical and Empirical Insights on the Jeffreys Test** *Accepted, 2026*
  Christophe Hurlin and Yoann Pull European Journal of Operational Research

Working Papers

- **Generalized Impulse Responses of Portfolio Default Probabilities: A Modular Framework with an Application to Geopolitical Risk** *2026*
  Guillaume Flament, Christophe Hurlin, Quentin Lajaunie and Yoann Pull SSRN: 7106918
  - Submitted to the *Journal of Applied Econometrics*.

- **E-calibration: Anytime-Valid Monitoring of Probability of Default Model Calibration** *2026*
  Yoann Pull Working paper

- **Reverse Stress Testing Geopolitical Risk in Corporate Credit Portfolios: A Formal and Operational Framework** *2026*
  Christophe Hurlin, Quentin Lajaunie and Yoann Pull arXiv preprint: 2601.03983

Conference Presentations and Seminars

- **18th Annual Society for Financial Econometrics Conference** *June 2026*
  Society for Financial Econometrics, University of Macau Macau, China

- **19th Financial Risks International Forum** *March 2026*
  Institut Louis Bachelier Paris, France

- **Research Seminar** *November 2025*
  Laboratoire d'Économie d'Orléans, University of Orléans Orléans, France

- **QFFE 2025 – Quantitative Finance and Financial Econometrics** *June 2025*
  Aix-Marseille School of Economics Marseille, France

- **ARAE – Association for Research in Applied Econometrics** *December 2024*
  Université Paris Dauphine-PSL Paris, France

Teaching

- **Climate Risk and Financial Risk Management** *2024 – present*
  ENSAI and University of Bordeaux Lecturer
  - Transmission channels from physical and transition climate risk to credit, market, and underwriting risk; implications for banking and insurance risk management, and for regulatory climate stress testing exercises.

- **Machine Learning: Support Vector Machines** *2025 – Present*
  M.Sc. in Econometrics and Applied Statistics (ESA), University of Orléans Lecturer
  - Support vector machines for classification and regression, primal and dual formulations, soft-margin SVM, kernel methods, and practical applications in Python.

- **Interpretable Machine Learning** *2025 – Present*
  M.Sc. in Econometrics and Applied Statistics (ESA), University of Orléans Lecturer
  - Model interpretability and explainability, model-agnostic methods, partial dependence plots, ICE, ALE, surrogate models, LIME, Shapley values, and SHAP.

- **Deep Learning for Time Series** *2026 – Present*
  M.Sc. in Financial Engineering, Paris Dauphine University – PSL Lecturer, forthcoming
  - Deep learning methods for time series modelling and forecasting, from recurrent and convolutional architectures to modern sequence models, attention-based methods, transformers, and recent implementations such as Kolmogorov–Arnold Networks.

- **Regularization Methods** *2026 – Present*
  M.Sc. in Applied Mathematics and Statistics, Financial Engineering Track, University of Bordeaux Lecturer, forthcoming
  - Regularization methods for statistical learning, including theoretical foundations, penalized regression, Ridge, Lasso, Elastic Net, model selection, sparsity, shrinkage, the bias–variance trade-off, and practical implementation.

Research and Professional Experience

- **Square Management** *March 2024 – September 2024*
  Quantitative Research Intern Paris, France
  - Integrated climate risk into asset evaluation models based on Fama–French factors.
  - Estimated the impact of climate risk at the company and sector levels using panel regressions.
  - Conducted literature reviews and drafted research material on density forecasting for Value-at-Risk and Expected Shortfall estimation.

- **Institut de la Statistique de la Polynésie Française (ISPF)** *May 2023 – August 2023*
  Data Scientist Intern — official statistical institute of French Polynesia Papeete, Tahiti, French Polynesia
  - Contributed to data processing work for the population census of French Polynesia.
  - Created OLAP cubes using SSAS and SQL to facilitate internal data usage.
  - Contributed to the implementation of machine learning methods for missing-data imputation.

Skills

- **Statistics and Econometrics:** Sequential testing, e-values, test martingales, high-dimensional statistics, Bayesian statistics, financial econometrics, credit risk modelling, probability of default calibration, stress testing, and model validation.
- **Programming:** Python, R, SQL, Git, Shiny.
- **Document Creation:** LaTeX, Markdown, Microsoft Office.
- **Languages:** French – native; English – C1.